

Bayesian multimodeling

MIPT

About the course

The focus of the course is on models, especially on complex compositions of models.

- Model criteria
- Model complexity
- Model search spaces
- What makes a model optimal or suboptimal?
- How do we optimize hyperparameters and metaparameters?
- What is the difference between hyperparameters and metaparameters?
- How do we combine models?
- How do we work with heterogeneous models?

Course overlap

- Bayesian model selection: a deeper course with a stronger focus on mathematical details
- Generative models: a course with a stronger focus on generative modeling

Topics for this term

- Distributions, expectation, and likelihood
- Bayesian inference
- Informative priors and sampling
- Minimum Description Length (MDL)
- Approximate inference
- Generative and discriminative models
- Graphical models
- Gaussian processes and their relation to state-space models

Scores

$\text{Score} = \min(10, \text{round}(2 + 2 \text{ Forms} + 6 \text{ Project} + \text{Publication bonus})) \circ \text{Paper review}$

- **Paper review**

- ▶ Every class, there will be 1–2 student talks.
- ▶ This year, the assignment is not only a talk but also a blog post describing the paper in detail.
- ▶ Before each talk, two students will be randomly selected as opponents. Each opponent must identify and critically assess the most significant weakness, limitation, or unresolved issue in the paper.
- ▶ Students who do not give at least one talk receive no score for the paper-review component.
- ▶ If a student gives only one talk, the maximum score is 7.

- **Forms** are **very** simple and are intended only as a fallback for the minimum student grade. ¹

¹Credits: Anastasia Voznyuk

Projects

Student projects are devoted to reproducing multiple well-established papers and packaging their implementations into an open-source library.

You can work in teams of up to 4 people. However, this means that the project must implement at least 4 different papers.

An approximate project schedule can be found on the course page.

Expected results from the team:

- An implemented open-source library
- A **short** blog post
- A technical report

For the brave ones, we can try to publish the results in a peer-reviewed journal (+1 point for all team members).

The list of projects and further details will be presented next week.

Examples of good project results:

- Relaxit: <https://github.com/intsystems/relaxit>
- bensemble: <https://github.com/intsystems/bensemble>

Talk

- **Timing:** 10–15 minutes
- **Structure:**
 - ▶ Title
 - ▶ Task and model motivation
 - ▶ Formal problem statement
 - ▶ Theory and method description
 - ▶ Experiments, examples, and applications
 - ▶ Literature
 - ▶ Two simple questions (will be included in the form)
- Poorly prepared talks receive a score of zero.

You also need to write a blog post explaining the paper in detail **for at least one talk**. The text should be written for a qualified audience: students in the 2nd–3rd year of our department.

Each blog post must be reviewed by another student, and the author must revise the post based on the review.

Homework

- Decide whether you are definitely attending the course.
- Fill in the form (the link will be available on the course page).
- Have a look at the libraries we might work with during the project:
 - ▶ JAX (and the JAX ecosystem)
 - ▶ Pyro
 - ▶ Julia (for enthusiasts)
- Check that you have access to the `intsystems` organization on GitHub.
- For those in charge: check that you are able to record classes and upload them to YouTube.

Page: <https://github.com/intsystems/BMM>
TG: see page :)